

NETWORKS AND COMPLEXITY

Solution 23-6

*This is an example solution from the forthcoming book Networks and Complexity.
Find more exercises at <https://github.com/NC-Book/NCB>*

Ex 23.6: Cat model [Lvl. 3]

A cat roams freely in the flat shown below. At rate 1 the cat becomes bored of the room she is currently in, then she picks one of the doors of the current room at random and walks through it. Determine the probability x_n that the cat is in room n in the stationary state.

Solution

Note that the cat does not follow our physical diffusion process. The differential equations are

$$\dot{x}_1 = -x_1 + x_2/2 + x_5/3 \quad (1)$$

$$\dot{x}_2 = -x_2 + x_1/2 + x_6/3 \quad (2)$$

$$\dot{x}_3 = -x_3 + x_6/3 \quad (3)$$

$$\dot{x}_4 = -x_4 + x_5/3 \quad (4)$$

$$\dot{x}_5 = -x_5 + x_1/2 + x_4 + x_6/3 \quad (5)$$

$$\dot{x}_6 = -x_6 + x_2/2 + x_3 + x_5/3 \quad (6)$$

So the Jacobian is

$$\mathbf{J} = \begin{pmatrix} -1 & 1/2 & 0 & 0 & 1/3 & 0 \\ 1/2 & -1 & 0 & 0 & 0 & 1/3 \\ 0 & 0 & -1 & 0 & 0 & 1/3 \\ 0 & 0 & 0 & -1 & 1/3 & 0 \\ 1/2 & 0 & 0 & 1 & -1 & 1/3 \\ 0 & 1/2 & 1 & 0 & 1/3 & -1 \end{pmatrix} \quad (7)$$

Because the number of walkers (1 cat) is conserved this is still a Laplacian matrix. (You can see that the columns sum to zero and all off-diagonal elements have the same sign, which is the characteristic we need.)

Now, how do we solve the system? Note that although the system has a cycle, our cat does not have a preference for walking clockwise or counter-clockwise around the flat. So we suspect that the system still has detailed balance in the stationary state. Let's use this.

The detailed balance condition between room 1 and 2 is

$$\frac{1}{2}x_1 = \frac{1}{2}x_2 \quad (8)$$

and hence

$$x_1 = x_2 \quad (9)$$

Between 2 and 6 we have the condition

$$\frac{1}{2}x_2 = \frac{1}{3}x_6 \quad (10)$$

so

$$x_6 = \frac{3}{2}x_2 = \frac{3}{2}x_1 \quad (11)$$

Between 6 and 3 we have

$$x_3 = \frac{1}{3}x_6 = \frac{1}{2}x_1 \quad (12)$$

Between 5 and 6 the condition is

$$\frac{1}{3}x_5 = \frac{1}{3}x_6 \quad (13)$$

and hence

$$x_5 = x_6 = \frac{3}{2}x_1 \quad (14)$$

If we were right about the detailed balance the condition between 5 and 1 should now be met automatically. Let's check

$$\frac{1}{3}x_5 = \frac{1}{2}x_1 \quad (15)$$

substituting our previous result reveals that this is true, so yes, the detailed balance assumption works out. Now we only have one room left to go. We consider the link between 4 and 5,

$$x_4 = \frac{1}{3}x_5 = \frac{1}{2}x_1 \quad (16)$$

The probabilities for the individual rooms should add up to 1, so

$$1 = \left(1 + 1 + \frac{1}{2} + \frac{1}{2} + \frac{3}{2} + \frac{3}{2}\right)x_1 = 6x_1 \quad (17)$$

which tells us

$$x_1 = x_2 = \frac{1}{6}, \quad x_3 = x_4 = \frac{1}{12}, \quad x_5 = x_6 = \frac{1}{4} \quad (18)$$

To verify these results you could substitute them into the differential equation or multiply the Jacobian with the vector $\mathbf{x}$.